Statistical Mechanics Homework (Chapter1)

Tang Yi

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题目 1. 1-1 假设在压强不太高时，某 1mol 的真实气体的物态方程为

$$pv = RT(1 + Bp)$$

其中，B 为温度的函数。求 α 和 κ，并给出在 $p \to 0$ 时的极限值。

题目 1 的解答. 由定义，得:

$$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p$$

代入，得

$$\begin{aligned}\alpha &= \frac{1}{v}\left(\frac{\partial v}{\partial T}\right)_p \\ &= \frac{1}{v}\left[\left(\frac{\partial v}{\partial T}\right)_{p,B} + \left(\frac{\partial v}{\partial B}\right)_{p,T}\left(\frac{\partial B}{\partial T}\right)_p\right] \\ &= \frac{1}{T} + \frac{pv}{1+Bp}\left(\frac{\partial B}{\partial T}\right)_p\end{aligned}$$

同理，对 κ，有

$$\kappa = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T$$

代入，得

$$\begin{aligned}\kappa &= -\frac{1}{v}\left(\frac{\partial v}{\partial p}\right)_T \\ &= -\frac{1}{v}\left(-\frac{RT}{p^2}\right) \\ &= \frac{1}{p(1+Bp)}\end{aligned}$$

当 $p \to 0$ 时，$\alpha \to \frac{1}{T}$，$\kappa \sim \frac{1}{p} \to \infty$。

题目 2. 1-2 假设

$$\alpha = \frac{v-a}{Tv},\ \kappa = \frac{3(v-a)}{4pv}$$

证明物态方程是

$$p^{3/4}(v-a) = AT$$

这里 a 和 T 均是常量。

题目 2 的解答. 由定义，不难得到

$$\mathrm{d}V = \alpha V \mathrm{d}T - \kappa V \mathrm{d}p$$

代入，有

$$\begin{aligned}
\mathrm{d}v &= \frac{v-a}{T}\mathrm{d}T - \frac{3(v-a)}{4p}\mathrm{d}p \\
\frac{\mathrm{d}v}{v-a} &= \frac{\mathrm{d}T}{T} - \frac{3}{4}\frac{\mathrm{d}p}{p} \\
\mathrm{d}(\ln(v-a)) &= \mathrm{d}\left(\ln T - \frac{3}{4}\ln p\right) \\
&= \mathrm{d}\left(\ln \frac{T}{p^{3/4}}\right) \\
p^{3/4}(v-a) &= AT
\end{aligned}$$

得证!

题目 3. 1-5 证明：

(a) $\left(\dfrac{\partial H}{\partial T}\right)_V = C_p\left(1-\dfrac{\alpha\mu}{\kappa}\right)$；

(b) $\left(\dfrac{\partial H}{\partial V}\right)_T = \dfrac{\mu C_p}{V\kappa}$；

(c) $\left(\dfrac{\partial T}{\partial V}\right)_H = \dfrac{\mu}{V(\mu\alpha-\kappa)}$。

题目 3 的解答. 我们规定一下记号：题目中的 μ 是焦耳-汤姆孙系数，即 $\mu = \left(\dfrac{\partial T}{\partial p}\right)_H$。

(a)

$$\begin{aligned}
\mathrm{RHS} &= C_p\left[1+\mu\frac{\left(\dfrac{\partial V}{\partial T}\right)_p}{\left(\dfrac{\partial V}{\partial p}\right)_T}\right]\\
&= C_p\left[1-\mu\left(\frac{\partial p}{\partial T}\right)_V\right]\\
&= \left(\frac{\partial H}{\partial T}\right)_p\left[1-\left(\frac{\partial p}{\partial T}\right)_V\left(\frac{\partial T}{\partial p}\right)_H\right]\\
&= \left(\frac{\partial H}{\partial T}\right)_p\left(\frac{\partial T}{\partial p}\right)_H\left[\left(\frac{\partial p}{\partial T}\right)_H-\left(\frac{\partial p}{\partial T}\right)_V\right]\\
&= -\left(\frac{\partial H}{\partial p}\right)_T\left(\frac{\partial T}{\partial p}\right)_H\left[\left(\frac{\partial p}{\partial T}\right)_H-\left(\frac{\partial p}{\partial T}\right)_V\right]\\
&= \left(\frac{\partial H}{\partial T}\right)_p+\left(\frac{\partial H}{\partial p}\right)_T\left(\frac{\partial p}{\partial T}\right)_V\\
&= \left(\frac{\partial H}{\partial T}\right)_V = \mathrm{LHS}
\end{aligned}$$

(b)

$$\begin{aligned}
\mathrm{RHS} &= \left(\frac{\partial T}{\partial p}\right)_H\left(\frac{\partial H}{\partial T}\right)_p\frac{1}{-\left(\dfrac{\partial V}{\partial p}\right)_T}\\
&= \left(\frac{\partial H}{\partial p}\right)_T\left(\frac{\partial p}{\partial V}\right)_T = \left(\frac{\partial H}{\partial V}\right)_T = \mathrm{LHS}
\end{aligned}$$

(c)

$$\begin{aligned}
\mathrm{RHS} &= \frac{\left(\dfrac{\partial T}{\partial p}\right)_H}{\left(\dfrac{\partial T}{\partial p}\right)_H\left(\dfrac{\partial V}{\partial T}\right)_p+\left(\dfrac{\partial V}{\partial p}\right)_T}\\
&= \frac{\left(\dfrac{\partial T}{\partial p}\right)_H}{\left(\dfrac{\partial V}{\partial p}\right)_H} = \left(\frac{\partial T}{\partial V}\right)_H = \mathrm{LHS}
\end{aligned}$$

题目 4. 1-6 (a) 证明焦耳系数为

$$\eta = \left(\frac{\partial T}{\partial V}\right)_U = \frac{1}{C_V}\left(p - \frac{T\alpha}{\kappa}\right)$$

(b) 证明焦耳-汤姆孙系数为

$$\mu = \left(\frac{\partial T}{\partial p}\right)_H = \frac{V}{C_p}\left(T\alpha - 1\right)$$

题目 4 的解答. (a)

$$\begin{aligned}
\mathrm{RHS} &= \left(\frac{\partial T}{\partial U}\right)_V \left[p + T\frac{\left(\frac{\partial V}{\partial T}\right)_p}{\left(\frac{\partial V}{\partial p}\right)_T}\right] \\
&= \left(\frac{\partial T}{\partial U}\right)_V \left[p - T\left(\frac{\partial p}{\partial T}\right)_V\right] \\
&= \left(\frac{\partial T}{\partial U}\right)_V \left[-\left(\frac{\partial U}{\partial V}\right)_S - \left(\frac{\partial U}{\partial S}\right)_V \left(\frac{\partial S}{\partial V}\right)_T\right] \\
&= -\left(\frac{\partial T}{\partial U}\right)_V \left(\frac{\partial U}{\partial V}\right)_T = \left(\frac{\partial T}{\partial V}\right)_U = \mathrm{LHS}
\end{aligned}$$

(b)

$$\begin{aligned}
\mathrm{RHS} &= \left(\frac{\partial T}{\partial H}\right)_p \left[T\left(\frac{\partial V}{\partial T}\right)_p - V\right] \\
&= \left(\frac{\partial T}{\partial H}\right)_p \left[-\left(\frac{\partial H}{\partial S}\right)_p \left(\frac{\partial S}{\partial p}\right)_T - \left(\frac{\partial H}{\partial p}\right)_S\right] \\
&= -\left(\frac{\partial T}{\partial H}\right)_p \left(\frac{\partial H}{\partial p}\right)_T = \left(\frac{\partial T}{\partial p}\right)_H = \mathrm{LHS}
\end{aligned}$$

题目 5. 1-8 某热力学系统，其热容为温度的函数：$C(T) = AT^3$，若取 $T = 0\mathrm{K}$ 时，熵 $S_0 = 0$。试证明温度为 T 时，$S(T) = \frac{1}{3}C(T)$。

题目 5 的解答.

$$\begin{aligned}
C &= \frac{\mathrm{d}Q}{\mathrm{d}T} = T\frac{\mathrm{d}S}{\mathrm{d}T} \\
&= \frac{\mathrm{d}S}{\mathrm{d}(\log T)} = A\mathrm{e}^{3\log T} \\
\Rightarrow S &= \frac{1}{3}AT^3 + const
\end{aligned}$$

当 $T = 0\mathrm{K}$ 时，熵 $S_0 = 0$，故 $const = 0$。则 $S(T) = \frac{1}{3}C(T)$。

题目 6. 1-13 体积分别为 V_1 和 V_2 的两个容器，都装有 N 个分子的理想气体，压强均为 p，但温度分别为 T_1 和 T_2。现把它们链接起来，就热接触和连通两种方式，计算新的平衡时整个系统的熵变。

题目 6 的解答. 对热接触而言，由于分子数相同，两容器的定容热容 C_V 是相同的，则有

$$\mathrm{d}U = C_V \mathrm{d}T = -p\mathrm{d}V + T\mathrm{d}S$$

对 V_1:

$$\mathrm{d}U_1 = \text{đ}Q_1 = C_V \mathrm{d}T_1$$

对 V_2:

$$\mathrm{d}U_2 = \text{đ}Q_2 = C_V \mathrm{d}T_2$$

同时由于 $\text{đ}Q_1 + \text{đ}Q_2 = 0$，则有:

$$\mathrm{d}T_1 + \mathrm{d}T_2 = 0$$

积分，同时设热平衡的温度为 T，则有:

$$T_1 + T_2 - 2T = 0$$

则熵变为:

$$\begin{aligned}\Delta S &= \int_{T_1}^{T} \frac{C_V \mathrm{d}T_1}{T_1} + \int_{T_2}^{T} \frac{C_V \mathrm{d}T_2}{T_2} \\ &= C_V \ln \frac{T}{T_1} + C_V \ln \frac{T}{T_2} \\ &= C_V \ln \frac{(T_1+T_2)^2}{4T_1T_2}\end{aligned}$$

对于连通的方式，不仅涉及到热平衡引起的熵增，还有气压平衡引起的熵增。热平衡的熵增前已计算出，下面计算气压平衡导致熵增。

假设此时系统已经热平衡，再打开阀门扩散，设平衡后的压强为 p'，则有:

$$p'(V_1 + V_2) = 2NkT$$

可解得:

$$p' = \frac{Nk(T_1+T_2)}{V_1+V_2}$$

则我们可认为，V_1 的气体体积从 V_1 等温变为为 $V_1' = \dfrac{NkT}{p'}$，V_2 的气体体积从 V_2 等温变为为 $V_2' = \dfrac{NkT}{p'}$。我们在这两个态之间构建可逆过程，则有:

$$\begin{aligned}\Delta S' &= \int_{V_1}^{V_1'} \frac{Nk}{V}\mathrm{d}V + \int_{V_2}^{V_2'} \frac{Nk}{V}\mathrm{d}V \\ &= Nk \ln \frac{V_1'}{V_1} + Nk \ln \frac{V_2'}{V_2} \\ &= Nk \ln \frac{(NkT)^2}{p'^2 V_1 V_2} \\ &= Nk \ln \frac{(V_1+V_2)^2}{4V_1V_2}\end{aligned}$$

则连通方式整个的熵变化为:

$$\Delta S_{\text{total}} = \Delta S + \Delta S' = C_V \ln \frac{(T_1 + T_2)^2}{4T_1T_2} + Nk \ln \frac{(V_1 + V_2)^2}{4V_1V_2}$$